

The Dynasty Genius Architecture: Valuation Normalization and Machine Learning Calibration in Sports Analytics

1. Executive Recommendation

The convergence of high-frequency sports tracking data, predictive machine learning modeling, and rigorous asset management principles has necessitated the development of a highly structured, deterministic knowledge architecture for dynasty fantasy football.¹ Within this environment, where long-term valuation shifts dynamically based on marginal efficiency gains and strict biological aging constraints, a reliance on subjective narrative or uncalibrated modeling constitutes a catastrophic failure of product design.¹ The Dynasty Genius architecture is currently navigating a critical inflection point in Phase 14: resolving the discontinuity between pre-professional prospect forecasting (Engine A) and active-player performance modeling (Engine B).²

Currently, the system utilizes two distinct predictive mechanisms operating on entirely different feature spaces and prediction horizons.² While these engines successfully isolate college production vectors from professional tracking data, their outputs must ultimately converge into a single, unified metric: the Dynasty Value Score (DVS).² The existing limitation—wherein active veterans promoted to Engine B display a null DVS—prevents the calculation of cross-positional Value Above Replacement (VAR), disables market divergence flags, and fractures the Unified Player Value Object (PVO).²

To resolve these architectural blockers and satisfy the requirements of Phase 14, the following executive recommendations are established:

1. **DVS Normalization (Engine A to Engine B):** The system must reject the direct application of Engine A's P90 constants to Engine B outputs. Because Engine B predicts a two-year average points per game (PPG) ($avg_ppg_{t_1:t_2}$), its variance is naturally compressed compared to Engine A's single-season peak target (y^{24_ppg}).² The recommended architecture adopts **Option C: Historical Training Outcome P90**, computing a bespoke set of $POSITION_P90_AVG_PPG_B$ constants derived directly from the active veteran training distribution.²
2. **Order of Operations (Within-Position vs. Cross-Position):** Valuation must proceed systematically. The architecture will compute within-position standardization first (utilizing Z-scores relative to positional replacement baselines), followed by cross-positional normalization (utilizing a VORP-based multiplier) to map raw points to a

unified scarcity-adjusted scale.³

3. **DVS Scale Transformation:** The user-facing Dynasty Value Score must transition from a compressed 0–100 scale to a **0–1000 scale**.⁵ This mitigates the necessity for fractional values, prevents excessive tie clusters in rank-ordering, and scales seamlessly with the E27 (2027 1st-Equivalents) currency math utilized in trade execution.⁵
4. **Replacement Baselines (12-Team Superflex PPR):** Statistical mapping of starting requirements dictates the following absolute replacement baselines for VAR calculations: **QB25, RB33, WR53, and TE13**.⁶
5. **Calibration Methodology:** The system will utilize a hybrid calibration approach. **Z-score standardization** will handle the parametric scaling of projected points relative to the baseline ³, while **Isotonic Regression** is reserved strictly for calibrating binary probability outputs (e.g., breakout probability or cliff-onset likelihood) to avoid the step-function tie-clustering that ruins ordinal player rankings.⁸
6. **Prospect-to-Veteran Bridge:** To handle the "Dead Window" across Years 1-3, the system will execute a Bayesian-style blended transition based on games played (*games_t*), utilizing the Engine A score as the prior and the Engine B score as the likelihood.² However, for the immediate Phase 14 deployment, players with insufficient data will retain their Engine A score with a mandatory, explicit display caveat indicating insufficient professional data.²

2. Evidence Table: Claims, Sources, and Implementation Implications

The following table synthesizes the empirical evidence, statistical methodologies, and analytical claims driving the Phase 14 architecture.

Analytical Claim / Data Point	Source(s)	Confidence Level	Phase 14 Implementation Implication
Engine A Target Definition: Engine A predicts a single-season peak outcome, specifically defined as <i>y24_ppg</i>	²	Ground Truth	Engine A P90 constants cannot be directly mapped to Engine B, as the variance of a single-season peak exceeds that of a 2-year

(average PPG over years 2–4).			average.
Engine B Target Definition: Engine B predicts a 2-year average PPG covering years T+1 and T+2 to reduce noise and align with dynasty value.	2	Ground Truth	A bespoke set of <i>P90</i> normalization constants must be computed from the engine_b_features_v2.csv distribution.
Isotonic Regression Constraints: Isotonic regression produces step-wise, piecewise constant functions that generate ties in output probabilities.	8	High	Isotonic regression cannot be used to calibrate the final continuous DVS, as tie clusters destroy the ordinal ranking necessary for the Rookie Board.
Z-Score Normalization: Z-scores provide a mathematically robust method for standardizing inputs across distributions with differing variances.	3	High	Within-position valuation must use Z-scores to measure standard deviations above the replacement baseline before applying cross-positional weights.
Replacement Baselines: In	6	High	These exact integers serve as

12-team Superflex PPR formats, replacement levels sit at approximately QB25, RB33, WR53, and TE13.			the $\mu_{replacement}$ thresholds for all Value Above Replacement (VAR) calculations.
Draft Capital Supremacy: NFL Draft Capital is the single most predictive variable in dynasty forecasting due to organizational commitment.	¹	Ground Truth	Engine A must serve as a heavily weighted prior during the Year 1-3 bridge, as draft capital remains predictive even amidst early NFL inefficiency.
Aging Curve Continua: Hardcoded age cliffs (e.g., RB 26, WR 28) are being replaced by continuous piecewise linear fitted aging curves in Engine B.	²	High	DVS normalization must account for biological decay dynamically, applying exponential discounting to assets surpassing peak age thresholds.
TE Promotion Constraint: TEs are promoted to ACTIVE_B but remain on an experimental v1 fallback model.	²	Ground Truth	The PVO must append the engine_b_experimental_v1_fallback caveat to all TE veterans, and DVS calculations must flag high uncertainty.
E27 Currency Conversion:	²	High	The normalized DVS scale must

Trade values are normalized to a 2027 Mid 1st (1.00 E27), with ratios driven by position, age, and health multipliers.			easily translate into E27 multipliers to execute automated trade evaluation.
Value Over Replacement (VORP): VORP translates normalized points into true market value by subtracting replacement-level baselines.	4	High	DVS cannot be a purely independent metric; it must be a function of VORP to reflect Superflex positional scarcity.

3. Recommended Dynasty Value Score (DVS) Architecture

The objective of Phase 14 is to architect a unified valuation layer that permits the simultaneous, mathematical evaluation of a 21-year-old collegiate prospect and a 28-year-old active professional.² This requires translating divergent machine learning outputs into a universal scale without violating the mathematical realities of their distinct prediction horizons.²

3.1 Resolving the Engine A vs. Engine B Normalization Gap

Engine A (Prospect Forecasting) relies on metrics with demonstrated predictive validity, such as College Dominator Rating (e.g., $> 30\%$ for elite RBs, $> 25\%$ for alpha WRs), Yards Per Route Run (YPRR > 2.5), Age at NFL Entry, and Draft Capital.¹ Its outcome variable is the expected peak production in the early career (y_{24_ppg}).² Engine B (Active Veteran Forecasting), conversely, explicitly bans pre-NFL data to prevent feature leakage, relying instead on professional efficiency (EPA/play, CPOE, target share) and continuous fitted aging curves to predict a two-year moving average ($avg_ppg_{t_1-t_2}$).²

A critical statistical phenomenon occurs between these two targets: regression to the mean.¹⁰ A distribution of single-season peaks will mathematically exhibit a wider variance and a higher 90th percentile (P90) ceiling than a distribution of two-year averages, as consecutive elite

seasons are rarer than isolated spikes.²

Currently, Engine A normalizes scores using established P90 constants: WR 12.7, RB 14.6, TE 9.1, and QB 16.7.² If these constants were applied to Engine B's smoothed two-year averages (Option A), veteran DVS values would be systematically understated, creating an artificial market bias toward unproven rookies.² Option B (Distribution Percentile Ranking) must also be rejected, as relative percentiles fluctuate based on the league environment rather than the intrinsic utility of the asset.²

Therefore, the architecture dictates **Option C: Historical Training Outcome P90**.² The system will query the `engine_b_features_v2.csv` data store to isolate the true historical P90 of the *avg_ppg_t1_t2* target variable for each position. The Engine B DVS will then be normalized against this bespoke veteran ceiling, mathematically aligning the *value* of an elite veteran average with the *value* of an elite prospect peak.²

3.2 Order of Operations: Within-Position vs. Cross-Position

A foundational question in sports analytics valuation is whether to normalize scores directly across all positions, or to standardize within-position first.³ A raw cross-positional division is flawed because it ignores the variable distribution widths of different positions.¹¹ For example, the difference in points per game between the QB1 and the QB12 is significantly smaller than the gap between the RB1 and the RB12.⁷

The prescribed architecture mandates a two-step sequence:

1. **Within-Position Standardization:** The raw projected PPG is first standardized into a Z-score relative to the position's replacement baseline.³ This establishes exactly how many standard deviations an asset sits above irrelevance.
2. **Cross-Position Normalization (VORP Integration):** The positional Z-score is then translated into a cross-positional magnitude by applying Value Over Replacement Player (VORP) multipliers.⁴ Because Superflex formats generate extreme positional scarcity for quarterbacks, a QB with a modest Z-score may still command a higher cross-positional DVS than a WR with a high Z-score due to positional supply and demand.⁷

3.3 The User-Facing Scale: Transitioning to 0–1000

While the existing prospect model utilizes a 0–100 scale clamped at 100², Phase 14 requires expanding the user-facing DVS to a **0–1000 scale**.⁵

The rationale is twofold:

1. **Tie Mitigation:** As the Player Value Object (PVO) population expands to include both rookies and hundreds of active veterans, a 0–100 scale forces excessive rounding, resulting in clustered ties.⁸ In asset management, a DVS of 84.1 must be distinguishable

from 84.8 to accurately sort the Rookie Board and calculate marginal trade deltas.²

2. **Trade Currency Alignment:** The system's automated trade execution logic relies on E27 (2027 1st-Equivalents) currency math.² A 2027 Mid 1st is anchored at 1.00 E27, an Early 1st at 1.50 E27, and a Late 2nd at 0.20 E27.² A 0–1000 DVS scale maps cleanly to these decimals. An elite asset with a 950 DVS can conceptually map to multi-E27 valuations (e.g., 3.00 E27) without the cognitive dissonance of fractions. It provides visual granularity that aligns with professional market aggregators.⁵

4. Calibration Methods: Mathematical Approaches and Efficacy

Translating machine learning outputs (whether classification probabilities or regression estimates) into decision-grade intelligence requires strict calibration.⁸ An uncalibrated model may accurately rank players but fail entirely at estimating the true magnitude of their performance.¹⁴ The following calibration frameworks have been evaluated for Phase 14.

4.1 Isotonic Regression

Isotonic regression is a non-parametric method utilized to calibrate model outputs by finding a non-decreasing, step-wise function that maps predicted scores to empirical probabilities.⁸ It utilizes the Pool-Adjacent-Violators (PAV) algorithm to correct instances where higher model scores falsely correlate with lower real-world outcomes.¹⁵

- **Pros:** It makes no assumptions about the underlying distribution (unlike Platt Scaling) and provides highly robust probability mapping for sparse, non-linear data.⁸
- **Cons:** The critical failure mode of isotonic regression in sports ranking systems is that it produces a piecewise constant function.⁸ This inherently introduces ties among subsets of predictions.⁸
- **Implication:** Isotonic regression is **rejected** as a method for calibrating the continuous Dynasty Value Score, as introducing ties destroys the strict ordinal ranking necessary for the Rookie Board and divergence flags.⁹ It is reserved exclusively for calibrating secondary binary classifiers, such as the probability of a player hitting their "age cliff" in a given season.¹⁶

4.2 Z-Score Standardization

The Z-score standardizes features by subtracting the population mean and dividing by the standard deviation ($Z = (x - \mu) / \sigma$).³

- **Pros:** Z-scores elegantly resolve the issue of cross-metric comparison.³ When evaluating disparate metrics—such as merging a WR's 28% Target Share with a 2.5 YPRR—Z-scores normalize the inputs.¹ Furthermore, they highlight the value of elite outliers by quantifying

exact standard deviations from the mean.³

- **Cons:** They are highly sensitive to extreme outliers and assume a relatively normal distribution.
- **Implication:** Z-scores are the **approved methodology** for the first stage of DVS assembly, standardizing predicted PPG relative to the positional replacement baseline.³

4.3 Value Over Replacement Player (VORP)

VORP is an econometric calculation that measures a player's raw point projection minus the projected points of the best available free agent or "worst starter" at that position.⁴

- **Pros:** VORP is the gold standard for defining cross-positional utility.⁴ In Superflex dynasty formats, quarterbacks score the most points, but they are also subject to extreme lineup scarcity (only 32 exist globally).¹² VORP accurately elevates scarce, high-leverage assets above abundant, easily replaceable assets.⁴
- **Cons:** VORP relies entirely on the accurate identification of the replacement baseline; if the baseline is misidentified, the entire valuation structure collapses.¹⁷
- **Implication:** VORP is the **approved mechanism** for translating within-position Z-scores into the final cross-positional DVS multiplier.⁴

4.4 Exponential Scaling

Exponential scaling applies a non-linear multiplier to top-percentile outputs, reflecting the reality that elite, difference-making assets provide compounding leverage in head-to-head match-ups.¹⁸

- **Implication:** Exponential scaling is natively handled by the E27 valuation multipliers defined in the trade logic.² A 23-year-old running back producing 18 PPG receives a 1.2x positional multiplier and a 1.5x age multiplier, exponentially increasing their required trade return.² This logic will run parallel to, but separate from, the base DVS algorithm.

5. Superflex Positional Replacement Baselines

To execute the VORP-based normalization of the DVS, the system must establish mathematically rigorous replacement-level baselines.¹⁷ The concept of "replacement level" dictates that value is only derived from points scored above the lowest-ranked player occupying a starting roster slot.⁴

The system architecture specifies a 12-Team, Superflex, Full PPR format.⁶ The standard starting requirements are: 1 QB, 2 RB, 3 WR, 1 TE, 1 Standard Flex (RB/WR/TE), and 1 Superflex (QB/RB/WR/TE).⁶

The calculation of the exact $\mu_{\text{replacement}}$ rank per position proceeds as follows:

5.1 Quarterback Baseline: QB25

In a Superflex format, the optimal game theory dictates starting a quarterback in the Superflex slot due to their high scoring floor and consistent volume.⁷

- **Lineup Demand:** 12 teams $\times$ 1 primary QB = 12.
- **Superflex Demand:** 12 teams $\times$ 1 Superflex QB = 12.
- **Total Demand:** 24 starting QBs.
- **Baseline:** The first replacement-level player on the bench or waiver wire is **QB25**.⁶

5.2 Running Back Baseline: RB33

- **Primary Demand:** 12 teams $\times$ 2 starting RBs = 24.
- **Flex Utilization:** The standard flex spot is typically split between RB and WR based on roster construction. Historical allocation models distribute approximately 40% of standard flex starts to RBs and 60% to WRs in full PPR formats.⁶
- **Total Demand:** $24 + (12 \times 0.40) = 28.8$. Accounting for bye weeks and the necessity of one immediate bench replacement, the baseline settles at **RB33**.⁶

5.3 Wide Receiver Baseline: WR53

Wide receivers possess the deepest starting requirements and the highest supply.⁷

- **Primary Demand:** 12 teams $\times$ 3 starting WRs = 36.
- **Flex Utilization:** 60% allocation of the standard flex adds ~ 7.2 starters.
- **Total Demand:** ~ 43 . With the necessity of deeper bench coverage due to higher variance and 3-WR sets, the established replacement tier extends to **WR53**.⁶

5.4 Tight End Baseline: TE13

Despite the inclusion of TE Premium modifiers in some leagues, the standard starting requirement remains rigid.⁶

- **Primary Demand:** 12 teams $\times$ 1 starting TE = 12.
- **Flex Utilization:** Minimal in non-premium formats.²⁰
- **Baseline:** The replacement level sits squarely at the top of the waiver wire: **TE13**.⁶

These discrete ranks (QB25, RB33, WR53, TE13) will serve as the query anchors against the engine_b_features_v2.csv distribution. The average PPG of the players occupying these ranks forms the baseline subtracted from the projected point totals to calculate VORP.²

6. The Prospect-to-Veteran Bridge (Years 1–3)

One of the most complex challenges in machine learning sports modeling is navigating the transition period between collegiate forecasting and professional track records.²¹ NFL Draft Capital is the strongest initial predictor of success due to organizational commitment.¹ A team that spends the 6th overall pick on a player will aggressively force opportunity onto that asset; a 4th-round pick receives no such grace.¹

However, once a player enters the NFL, their first-year snap and route participation rapidly supersede draft capital as the clearest leading indicator of dynasty trajectory.¹ Players seeing $< 40\%$ route participation in Year 1 have historically devastating failure rates.¹

The transition logic dictates when a player stops routing through Engine A (Draft Capital/Collegiate Metrics) and begins routing through Engine B (NFL Tracking Data).²

6.1 The "Dead Window"

A systemic vulnerability occurs when a rookie completes their first season. Technically, they are no longer a "prospect".² However, they may lack the minimum threshold of professional games played or route-participation data required to generate a statistically significant Engine B projection.² If forcefully routed to Engine B, the model will output a null or heavily penalized score, improperly degrading a highly drafted player to a PRE_MODEL state.²

6.2 Bayesian Blending over Games Played

The mathematically optimal solution for bridging the gap across Years 1–3 is a Bayesian updating function.² Engine A's prospect score serves as the *Prior Probability*, while Engine B's professional tracking data serves as the *Likelihood*.²

The final DVS is a weighted average of the two engines, where the weight assigned to Engine B (W_B) is a sigmoid or linear function of professional games played ($games_t$).²

- At $games_t = 0$: $W_B = 0\%$ (Pure Engine A).
- At $games_t = 8$: $W_B = 50\%$ (Blended evaluation of draft pedigree and early usage).
- At $games_t = 17$: $W_B = 100\%$ (Pure Engine B; NFL usage fully overrides collegiate pedigree).

6.3 Recommended Phase 14 Implementation (Explicit Caveat Protocol)

While the Bayesian blending model is the ultimate North Star, Phase 14 requires immediate stabilization of the unified PVO.² The mandated implementation is **Option B: Explicit Caveat**

Tagging.²

Players in the Dead Window (defined as players in Years 1-2 who do not meet the minimum games_t threshold for Engine B) will continue to output their Engine A score as the baseline prior.² However, the architecture strictly enforces the Anti-Speed Protocol and transparency mandates.¹ Any player utilizing this fallback must have their PVO flagged with the mandatory display caveat: *"Insufficient professional season data — Engine A prospect score used as prior"*.² This ensures the end-user is acutely aware that the valuation rests on draft capital rather than verified professional efficiency.

7. Failure Modes and Governance Risks

The deployment of Phase 14 introduces several architectural and analytical vulnerabilities that must be actively monitored.

7.1 Data Leakage and Market Contamination

The most severe failure mode in dynasty forecasting is predicting the market rather than predicting the player.² Market aggregators like KeepTradeCut (KTC) and community consensus ADPs reflect human psychology, recency bias, and narrative hype.¹ If KTC values, consensus tiers, or ADP are allowed to leak into the Engine B feature matrix or the DVS normalization math, the model becomes a self-fulfilling feedback loop.²

- **Governance Action:** Strict enforcement of the Market Overlay constraint. Market values must only join the PVO *after* DVS scoring is complete.²

7.2 Miscalibration of the Divergence Engine (NOISE_BAND)

By populating DVS for all active veterans, the system will instantly activate Market Divergence flags across the entire NFL player pool.² The Divergence Engine fires when the delta between the model's DVS rank and the market's KTC rank exceeds a set threshold.² If the *NOISE_BAND* is calibrated too narrowly, the system will flood the user with false-positive "Buy/Sell" indicators triggered by minor, insignificant variance in the P90 normalization.²

- **Governance Action:** Divergence flags for veterans must remain suppressed or heavily guarded until the scheduled NOISE_BAND recalibration window closes in mid-July 2026.²

7.3 Binary Application of Age Cliffs

Historically, dynasty managers treat age cliffs (RB 26, WR 28, TE 30, QB 33) as absolute binary death sentences, selling elite assets for pennies on the dollar.¹ If the normalization logic hardcodes a sudden DVS penalty at these exact ages, it violates the architectural mandate of Engine B, which utilizes continuous, fitted piecewise linear models to evaluate biological decay

organically.² Elite athletes with minimal injury histories (e.g., high-RAS players) decay slower than volume-dependent grinders.¹

- **Governance Action:** Ensure the DVS normalization math consumes the raw output of the age-curve state natively, rather than applying a secondary hardcoded penalty layer.²

8. Explicit Out-of-Scope Items

To maintain engineering velocity and adhere to the product constitution, the following items are strictly out-of-scope for the Phase 14 deployment²:

1. **Tight End (TE) Model Refactoring:** TEs were promoted to ACTIVE_B in Phase 13.3, but they remain on an experimental v1 fallback model due to validation gate failures.² Attempting to rebuild the TE feature matrix or demote TEs back to PRE_MODEL status is out of scope.² The system will simply append the engine_b_experimental_v1_fallback caveat and proceed with DVS normalization.²
2. **Market Feature Modeling:** Incorporating KTC, FantasyCalc, or DynastyNerds ADP directly into the prediction algorithm remains permanently banned.²
3. **UI/UX Polish:** Deep formatting, color-coding, or aesthetic enhancements to the Trust Surface and Trade Lab are deferred until the mathematical calibration is fully trusted and validated.²
4. **Redraft Valuation:** The framework is built on a 3-7 year multi-season horizon.¹ Attempting to generate a secondary DVS specifically tailored for single-season redraft utility is a category error and is out of scope.¹

9. Open Decisions for the Executive Steering Committee

Before the engineering team commits to the Phase 14 implementation scripts, the following strategic decisions require explicit approval from the Executive Steering Committee (David)²:

1. **DVS Scale Transition Approval:** Approval is required to officially transition the user-facing DVS from a 0–100 clamped scale to a 0–1000 scale to improve sorting granularity and align with E27 trade math.
2. **VAR Threshold Lock:** Confirmation that the replacement baselines derived in Section 5 (QB25, RB33, WR53, TE13) are accurate representations of the user's specific league context, and approval to lock these integers as the $\mu_{replacement}$ query bounds.
3. **Bridge Logic Approval:** Approval of "Option B: Explicit Caveat Tagging" as the immediate bridge mechanism for the Dead Window, formally deferring the complex Bayesian blended-weighting model to a future phase.
4. **Divergence Engine Activation:** A decision on whether to permit veteran market-divergence flags to fire immediately upon DVS normalization, or to enforce a hard

lock until the mid-July 2026 noise-band calibration is complete.

10. Proposed Phase 14 Implementation Workstreams and Ordering

Upon approval of the open decisions, Phase 14 will execute via the following sequential workstreams ²:

Workstream 14.1: Empirical Data Extraction and Constant Generation

- *Task*: Run descriptive statistical queries on app/data/training/engine_b_features_v2.csv.²
- *Task*: Isolate the $avg_ppg_t_{1_t2}$ distribution for each position and extract the true 90th percentile bounds.²
- *Task*: Hardcode the new $POSITION_P90_AVG_PPG_B$ constants into the configuration layer.²

Workstream 14.2: DVS Normalization and PVO Assembly

- *Task*: Modify pvo_assembler.py to remove the line 316 block that currently forces dynasty_value_score = None for ACTIVE_B players.²
- *Task*: Implement the Z-score and VORP normalization algorithms to map the raw Engine B projections to the new 0–1000 DVS scale.²
- *Task*: Ensure the engine_b_experimental_v1_fallback caveat is correctly appended to all active Tight Ends.²

Workstream 14.3: The Prospect-to-Veteran Bridge

- *Task*: Implement the transition logic querying games_t.²
- *Task*: Build the fallback routing for players in the Dead Window, injecting the mandatory "Insufficient professional season data — Engine A prospect score used as prior" text into the caveat array.²
- *Task*: Execute an automated test against the 2024 and 2025 draft classes to verify that the canonical_player_id handoff from Sleeper to GSIS is intact.²

Workstream 14.4: VAR Activation and System Integration

- *Task*: Map the QB25, RB33, WR53, and TE13 baselines to the VAR calculation engine.²
- *Task*: Activate value_above_replacement generation for all Engine B players.²
- *Task*: Update the Rookie Board sorting logic to safely ingest and rank a mixed population of prospects and veterans utilizing the new 0–1000 DVS integers.²

This structured approach ensures that the Dynasty Genius architecture transitions from fragmented analysis into a mathematically rigorous, fully calibrated asset management system,

equipped to leverage long-term deterministic forecasting against the volatile sentiment of the dynasty market.

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